



Exp :12 Write the python program for Tic Tac Toe game

Aim:

To write a Python program to implement a two-player Tic Tac Toe game using functions and lists.

Algorithm:

Step 1: Create a list of 9 elements containing numbers '1' to '9' to represent the 3x3 Tic Tac Toe board.

Step 2: Define a function print_board to display the board in a 3x3 grid format with rows and columns separated by lines.

Step 3: Define a function check_win to check if a player has won by comparing all possible winning combinations (3 rows, 3 columns, and 2 diagonals).

Step 4: Define a function check_tie to check if the game is a draw by verifying if all 9 positions are filled with 'X' or 'O'.

Step 5: In the main function, initialize a variable current_player to 'X' and start a loop that runs until the game ends.

Step 6: Display the board and prompt the current_player to choose a position from 1 to 9.

Step 7: Validate the input to ensure the position is between 1 and 9 and is not already occupied.

Step 8: Update the board at the chosen position with the current_player's symbol ('X' or 'O').

Step 9: Check if the current_player has won using check_win. If yes, print the winner and stop the game.

Step 10: Check if the game is a tie using check_tie. If yes, print a tie message and stop the game.

Step 11: If neither condition is met, switch current_player from 'X' to 'O' (or vice versa) and repeat from Step 6.

Program:

```
def print_board(board):  
    print("\n " + board[0] + " | " + board[1] + " | " + board[2])  
    print("---|---|---")
```

```
print(" " + board[3] + " | " + board[4] + " | " + board[5])
print("---|---|---")
print(" " + board[6] + " | " + board[7] + " | " + board[8] + "\n")
```

```
def check_win(board, player):
```

```
    win_conditions = [
        [0, 1, 2], [3, 4, 5], [6, 7, 8], # Rows
        [0, 3, 6], [1, 4, 7], [2, 5, 8], # Columns
        [0, 4, 8], [2, 4, 6]             # Diagonals
    ]
```

```
    for condition in win_conditions:
```

```
        if board[condition[0]] == board[condition[1]] == board[condition[2]] == player:
            return True
```

```
    return False
```

```
def check_tie(board):
```

```
    for spot in board:
```

```
        if spot.isdigit():
            return False
```

```
    return True
```

```
def tic_tac_toe():
```

```
    board = ['1', '2', '3', '4', '5', '6', '7', '8', '9']
```

```
    current_player = 'X'
```

```
    game_over = False
```

```
    while not game_over:
```

```
        print_board(board)
```

```
        move = input(f"Player {current_player}, choose a position (1-9): ")
```

```
        if move.isdigit() and int(move) in range(1, 10):
```

```

index = int(move) - 1
if board[index].isdigit():
    board[index] = current_player

    if check_win(board, current_player):
        print_board(board)
        print(f"Player {current_player} wins!")
        game_over = True
    elif check_tie(board):
        print_board(board)
        print("It's a tie!")
        game_over = True
    else:
        current_player = 'O' if current_player == 'X' else 'X'
else:
    print("Position already taken! Try again.")
else:
    print("Invalid input! Please enter a number between 1 and 9.")

```

tic_tac_toe()

OUTPUT:

```

 1 | 2 | 3
---|---|---
 4 | 5 | 6
---|---|---
 7 | 8 | 9

Player X, choose a position (1-9): 1

 X | 2 | 3
---|---|---
 4 | 5 | 6
---|---|---
 7 | 8 | 9

Player O, choose a position (1-9): 5

 X | 2 | 3
---|---|---
 4 | O | 6
---|---|---
 7 | 8 | 9

Player X, choose a position (1-9): 9

```

Result: Thus, the program was executed and found that the output was found to be correct.